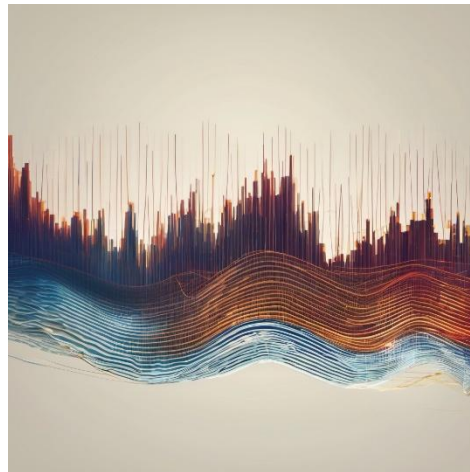


Final Presentation

Speech Controlled Brick Building

Supervisor: Marcel Ritter, PhD, BSc and Pascal Knierim, Ass.-Prof. Dr
Presentation: Hannes Röd



Created via <https://pixlr.com>

Motivation

- Speech controlled brick building
 - Speech as input



[14]



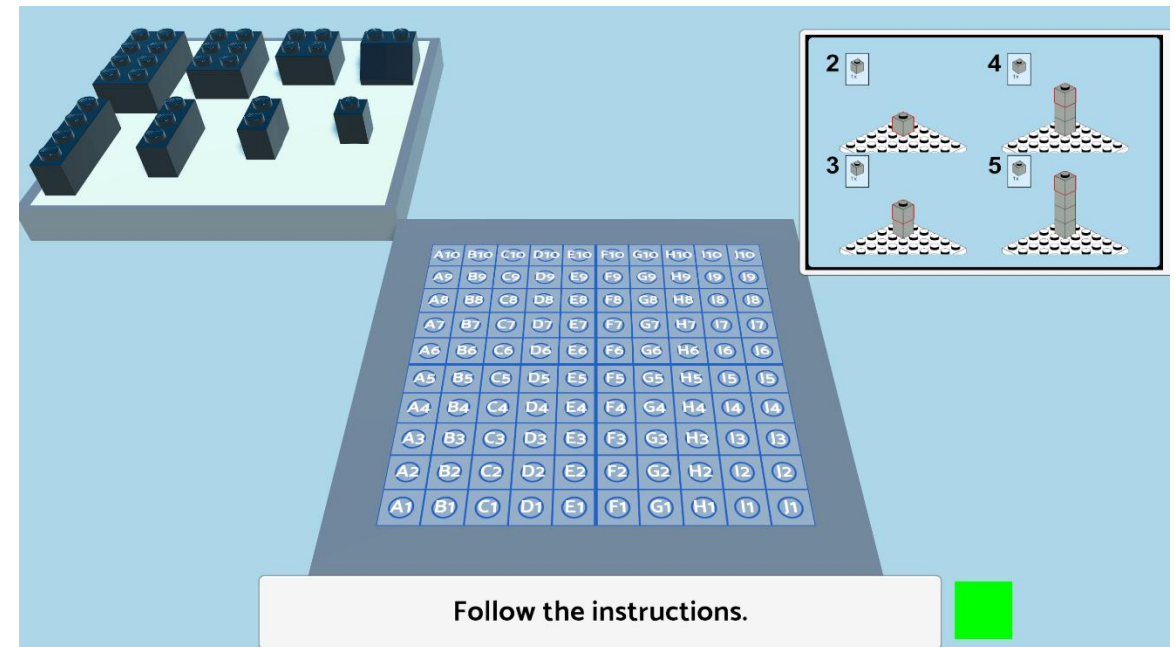
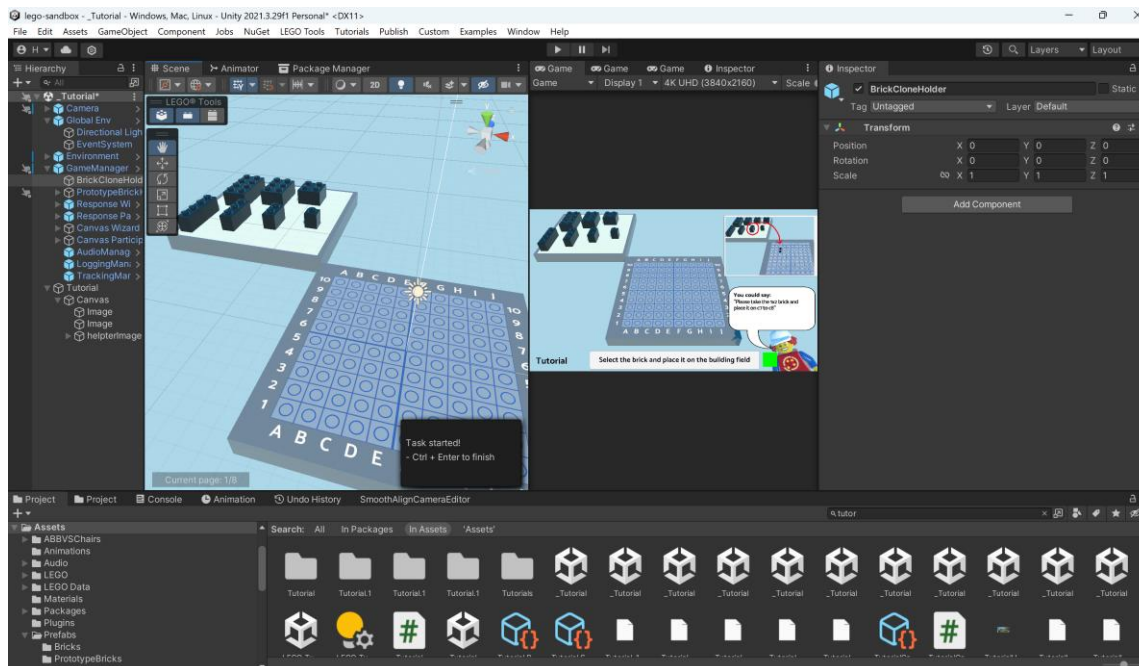
[10]



[4]

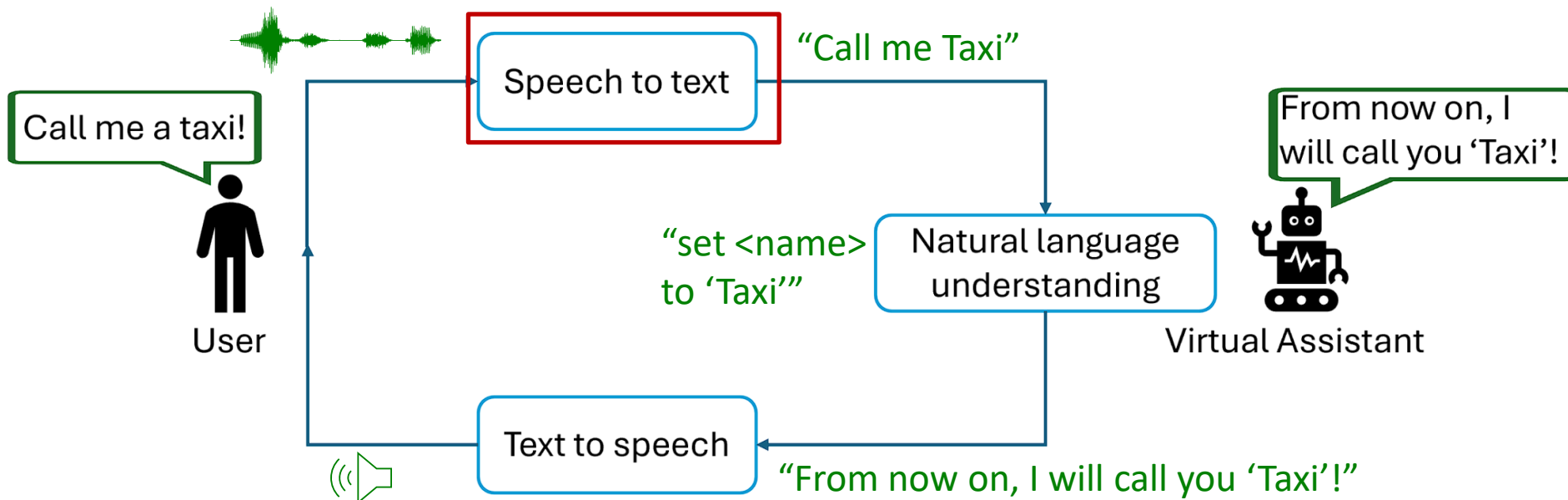
Motivation

- Speech controlled brick building
 - Speech as input
 - Integration into Unity



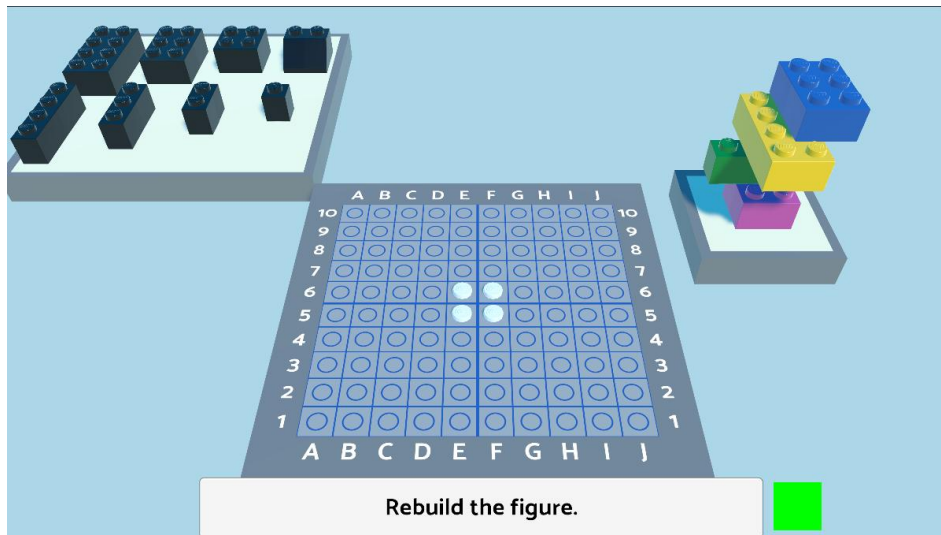
Motivation

- Automatic speech recognition (ASR)
 - Offline and open-source
 - Real-time solution
 - Benchmark using public dataset



Motivation

- User study
 - Evaluate building strategies
 - Refine environment
 - Collect data for two corpora



Audio-to-transcript



Command-to action

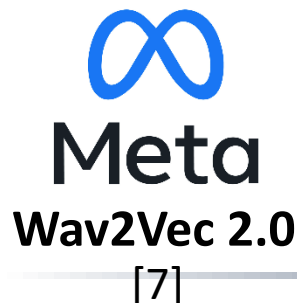
```
"playerId": 1,  
"logId": 5,  
"timestamp": "00:02:08.7321535",  
"audio_filepath": "/Audio/1/Task 3.1",  
"BrickActions": [  
  {  
    "action_type": "BrickAction",  
    "obj": "3005",  
    "brickId": "3005",  
    "action": "move",  
    "fromPosition": "(1.20, 2.90, 1.20)",  
    "toPosition": "(3.60, 1.93, 1.20)",  
    [...]
```

Overview

- Model Selection
- ASR
- Performance Metric
- Common Voice Benchmark
- User Study
- Evaluation
- Conclusion & Future Work

Model selection

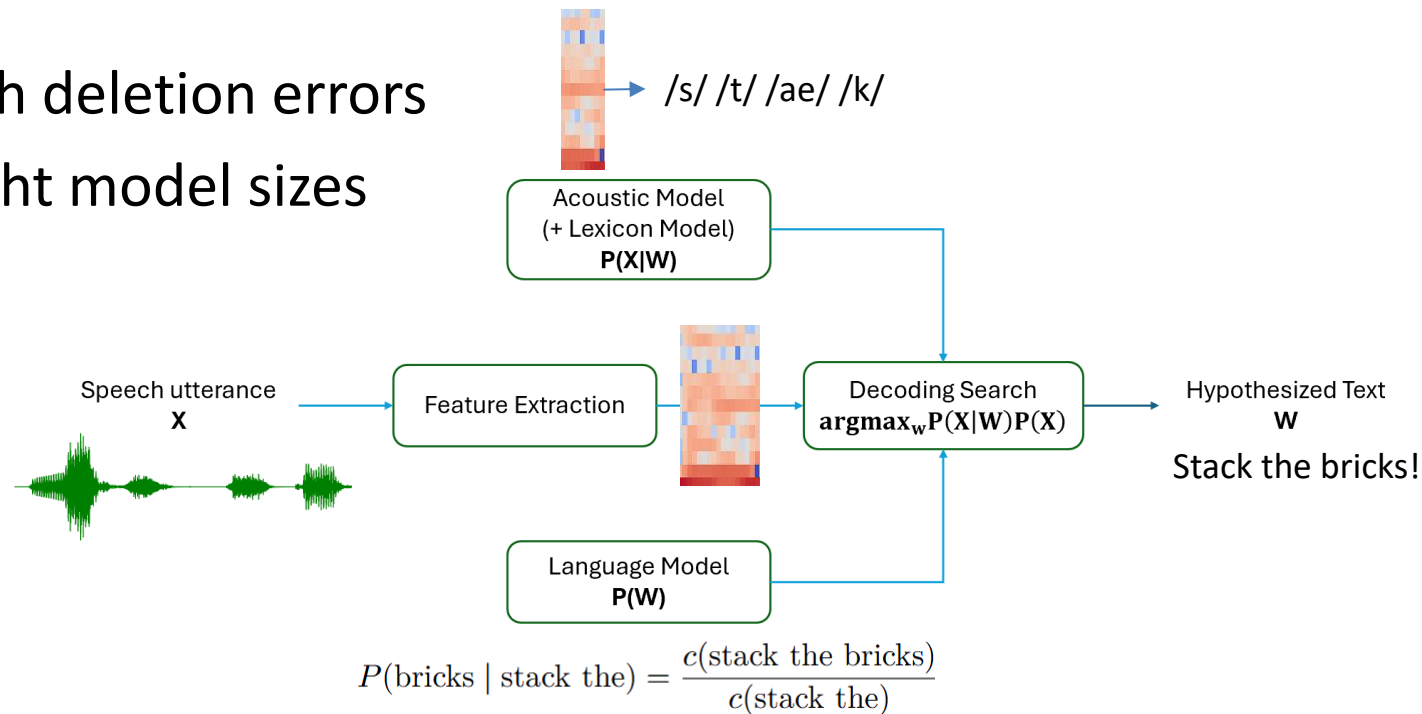
- Choosing the best candidate
 - Offline, open-source and trainable
 - English/German
 - Real-time
 - Vosk, Whisper, Wav2Vec 2.0



Toolkit/Framework	Model Family	Description	Approach	Reference
Windows Speech Recognition	-	Microsoft's built-in speech recognition engine for Windows	Hybrid HMM/GMM	[56]
Deep Speech	Mozilla Deep Speech	Model based on Baidu's research; popularized the concept of end-to-end speech recognition models	RNN and CTC loss	Mozilla, 2014 [30]
	Deep Speech 2	An improved version of Deep Speech with better accuracy and speed	RNN and CTC loss	Baidu, 2016 [5]
	PaddleSpeech (DeepSpeech)	Speech toolkit including self-supervised learning and streaming ASR	RNN; improvements on data processing	PaddleSpeech, 2022 [62]
Whisper	OpenAI Whisper	A general-purpose ASR model from OpenAI	Transformer	OpenAI, 2022 [50]
	Whisper C++	Whisper implementation in C++ for faster inference	Optimized for speed; same transformer architecture	[26]
	Whisper TFLite	TensorFlow Lite implementation of Whisper	Lightweight version for mobile and edge devices	[55]
Vosk	-	Toolkit based on Kaldi; provides integration to many platforms	GMM-HMM and DNN-HMM models	[34]
Nvidia NeMo	-	Speech recognition and natural language processing toolkit	RNN-T, transformer, and others	Nvidia NeMo, 2019 [41]
Nvidia OpenSeq2Seq	-	Toolkit for end-to-end ASR and TTS; predecessor of Nvidia NeMo	Transducer, CTC, and transformer	Nvidia OpenSeq2Seq, 2019 [40]
Meta Flashlight	Flashlight (wave2letter++)	Fast and flexible ML and DL in C++	NN models	Meta, 2022 [37]
Meta Wav2Vec 2.0	-	Self-supervised learning for SR	Transformer and CTC loss; self-supervised learning	Meta, 2020 [8]
	XLS-R	Multilingual extension of Wav2Vec 2.0, supporting 128 languages	Same architecture as Wav2Vec; cross-lingual self-supervised learning	Meta, 2021 [7]
Meta	SeamlessM4T	Multilingual and multitask model capable of translation and transcription	Transformer and multi-modal translation models for SR, TTS, and speech-to-speech	Meta, 2023 [16]
Kaldi	-	SR toolkit written in C++	HMM-GMM and DNN-HMM systems	Kaldi, 2011 [49]
SpeechBrain	-	PyTorch toolkit that accelerates conversational system development	Transducer, CTC, transformer	SpeechBrain, 2021 [51]
Sherpa	-	PyTorch framework focusing on end-to-end ASR	Transducer and CTC	Sherpa, 2023 [14]

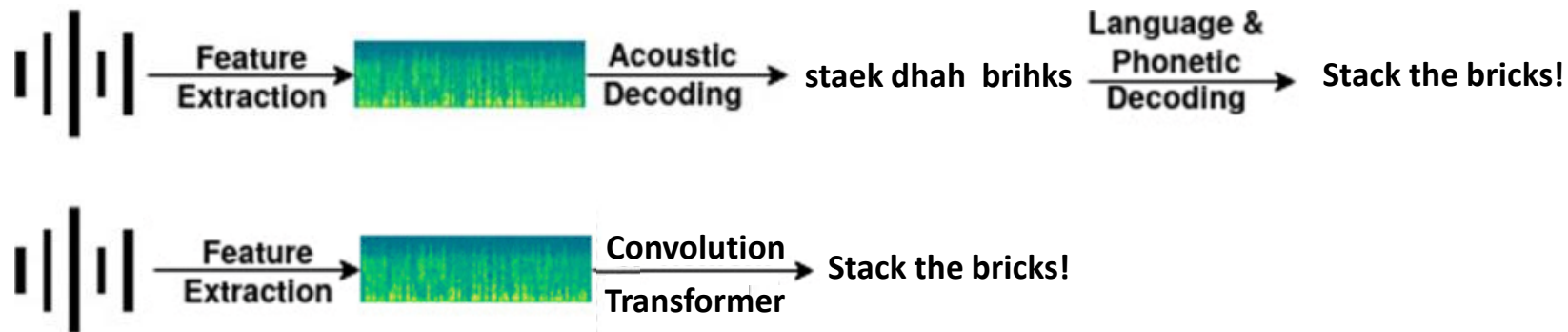
Overview of Speech Recognition Technology (1)

- Hybrid DNN-HMM approach
 - Pros: Separate, customizable models
 - Cons: High deletion errors
 - Lightweight model sizes



Overview of Speech Recognition Technology (2)

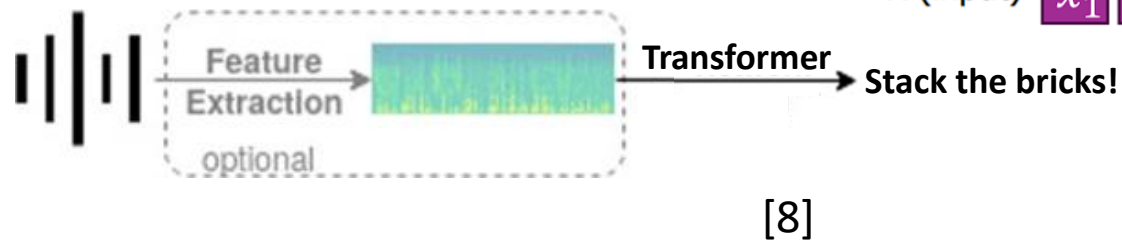
- End-to-End transformer approach
 - Pros: High accuracy
 - Cons: Slower inference
 - Less human intervention, but more training data needed



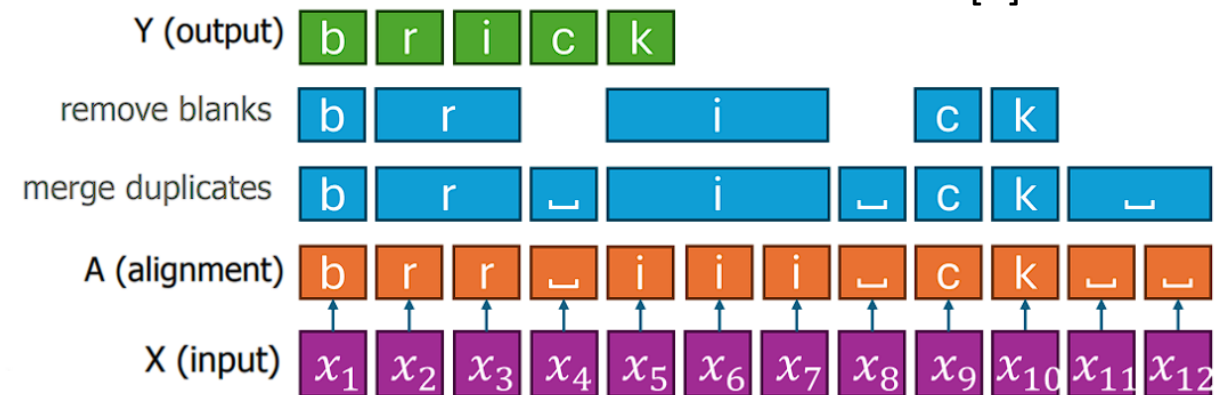
[8]

Overview of Speech Recognition Technology (3)

- Transformer + CTC approach
 - Pros: Fast inference + self-supervised learning (less labeled training data needed)
 - Cons: Lower accuracy without additional LM
 - Supports streaming




Meta
Wav2Vec 2.0
[7]



Performance Evaluation

■ Evaluation metric

— Word error rate (WER)

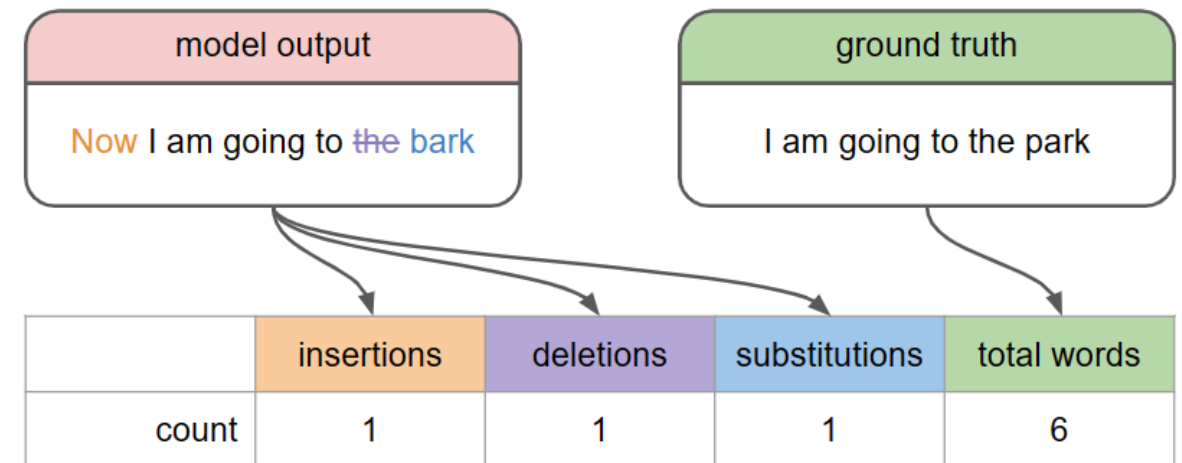
- Substitutions, deletions and insertions divided by #words

— Real time factor (RTF)

- Time to transcribe audio input divided by actual audio duration

$$WER = \frac{S + D + I}{N} \quad [1]$$

$$RTF = \frac{\text{Total Processing Time}}{\text{Total Duration}} \quad [1]$$



$$\text{word error rate} = \frac{1 + 1 + 1}{6} = 0.5 = 50\% \quad [12]$$

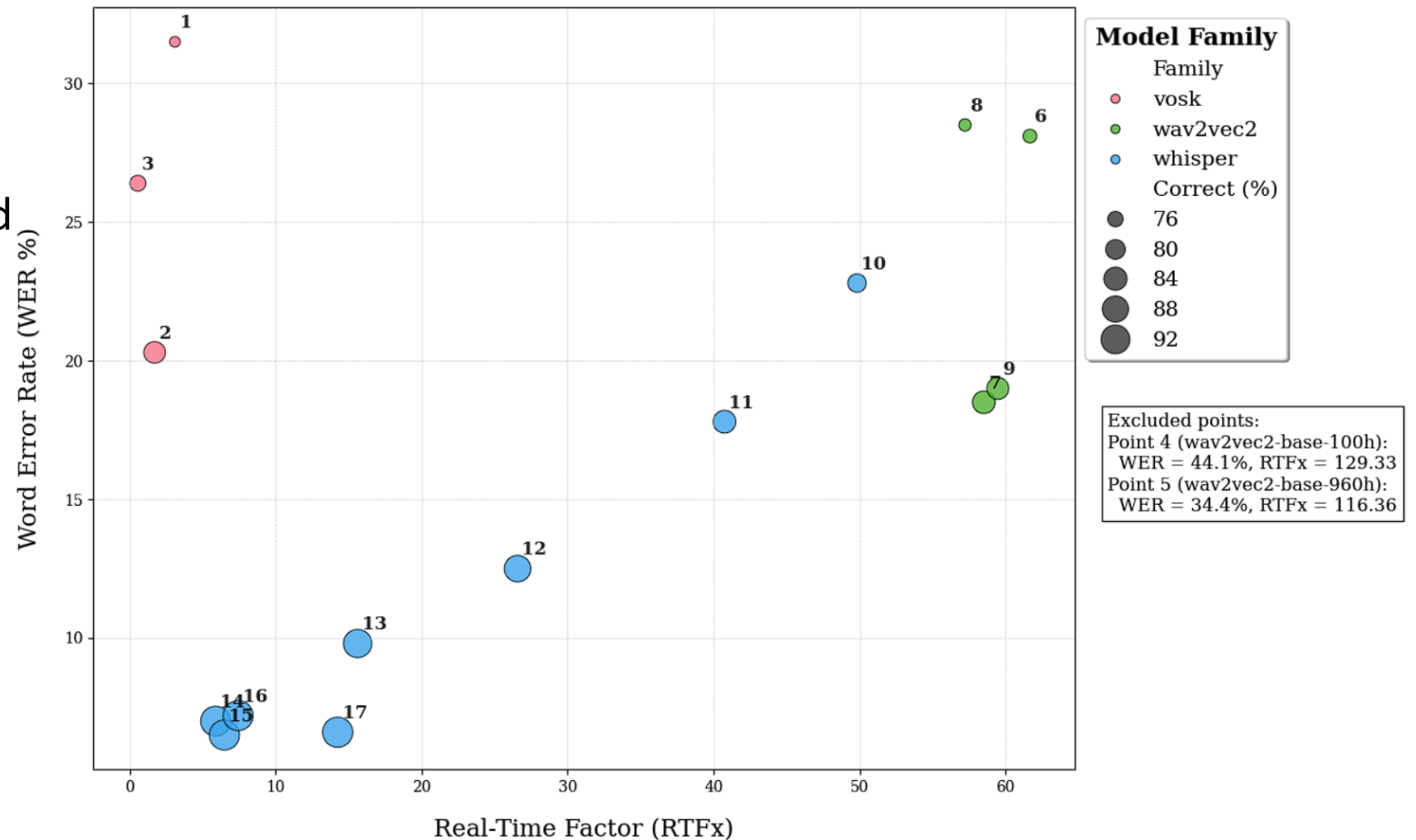
Common Voice Benchmark

- Delta Segment 17.0
- 18.18 hours of English audio
- 11000 validated samples

	Model	Subst. (%)	Del. (%)	Ins. (%)	WER (%)	SER (%)	RTFx
1	vosk-model-small-en-us-0.15	20.5	6.8	4.2	31.5	76.2	3.09
2	vosk-model-en-us-0.22	13.9	3.9	2.5	20.3	61.7	1.70
3	vosk-model-en-us-0.22-lgraph	17.6	5.9	2.9	26.4	72.9	0.55
4	wav2vec2-base-100h	35.0	5.5	3.6	44.1	89.7	129.33
5	wav2vec2-base-960h	27.1	3.7	3.6	34.4	82.8	116.36
6	wav2vec2-large-960h	22.1	3.2	2.8	28.1	77.5	61.68
7	wav2vec2-large-960h-lv60-self	14.5	2.2	1.9	18.5	65.2	58.52
8	wav2vec2-large-robust-ft-swbd-300h	22.9	3.4	2.2	28.5	78.9	57.23
9	wav2vec2-large-robust-ft-libri-960h	15.3	2.1	1.6	19.0	68.1	59.48
10	whisper-tiny.en	16.0	5.2	1.6	22.8	66.9	49.83
11	whisper-small.en	12.3	4.2	1.3	17.8	58.9	40.75
12	whisper-base.en	9.0	2.4	1.0	12.5	49.7	26.57
13	whisper-medium.en	7.4	1.6	0.9	9.8	44.3	15.61
14	whisper-large	5.7	0.8	0.6	7.0	37.0	5.88
15	whisper-large-v2	5.2	0.7	0.6	6.5	36.2	6.48
16	whisper-large-v3	5.7	0.8	0.7	7.2	38.5	7.42
17	whisper-large-v3-turbo	5.4	0.7	0.6	6.6	37.1	14.24
Global Best		—	—	—	6.5	36.2	129.33

Common Voice Benchmark

- Google Colab
 - Tesla T4 TPU with 15Gb VRAM and an Intel Xeon CPU with 2 virtual CPUs and 13GB of RAM



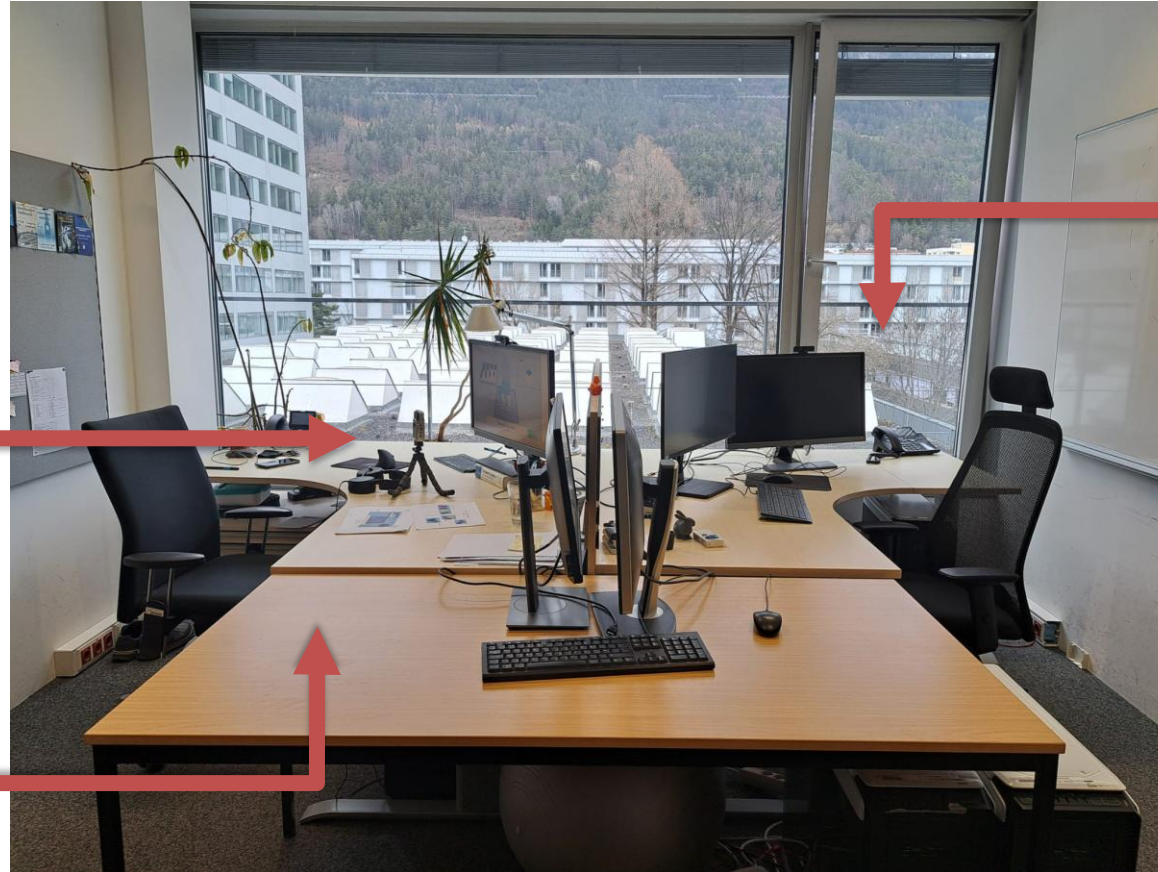
User Study

■ Setup

Microphone, monitor
and headphones

Participant

Operator

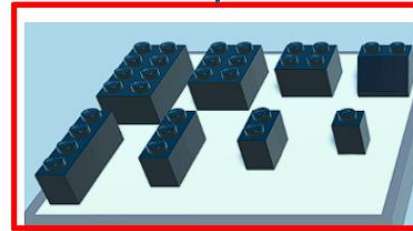


User Study

■ Environment

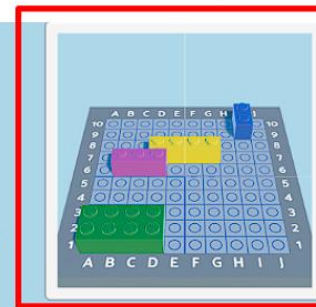
Prototype Bricks

All the bricks you can use during the building process



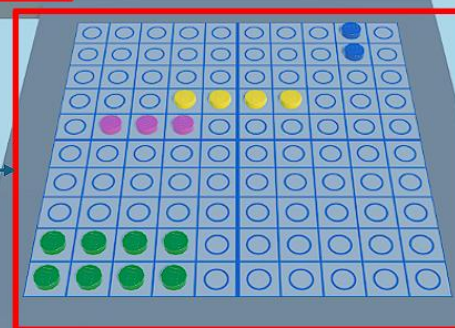
Instruction Manual

Building instructions get displayed in the top right corner



Building Field

Bricks can be placed here



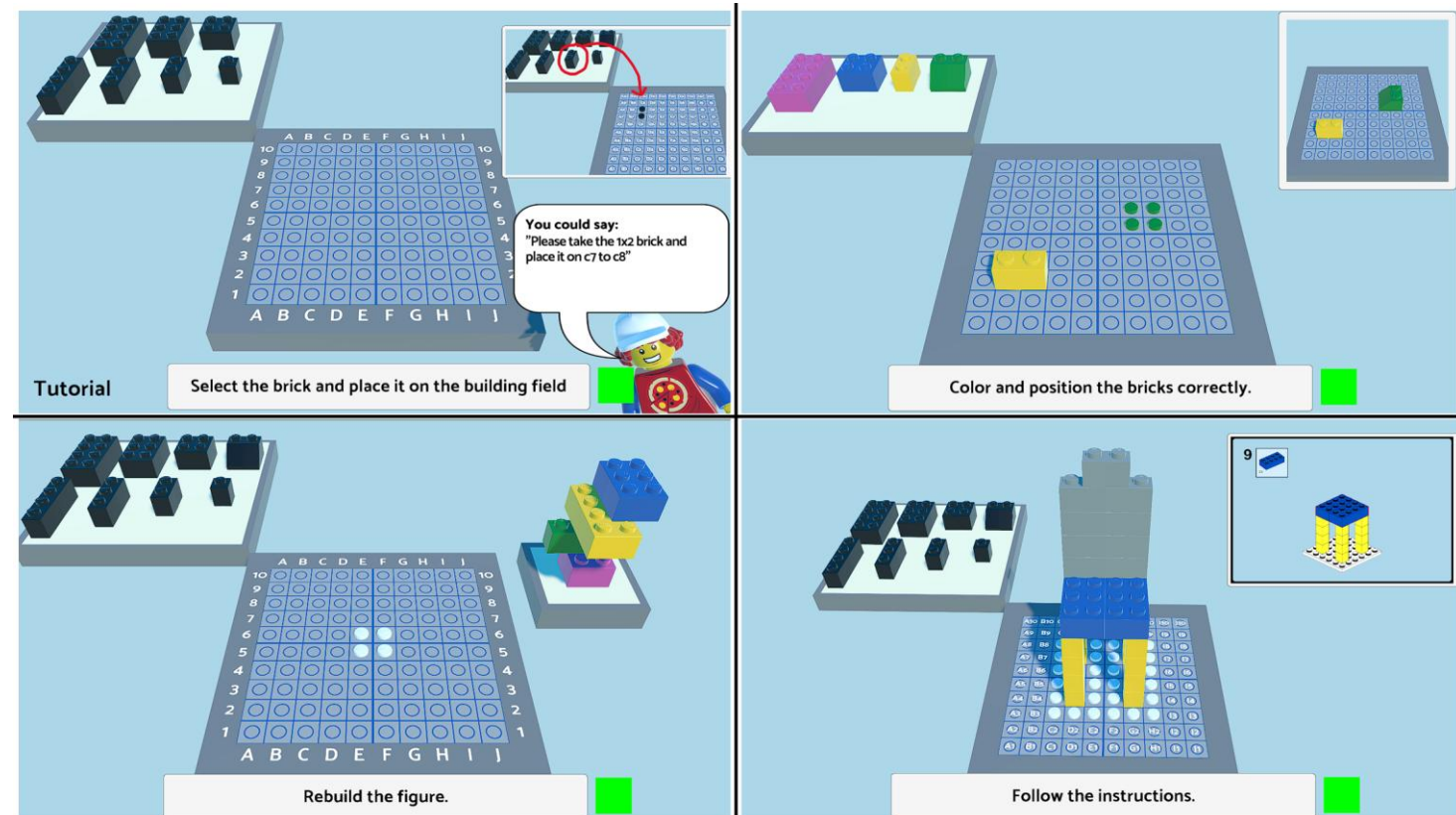
Color and position the bricks correctly.

Information Box

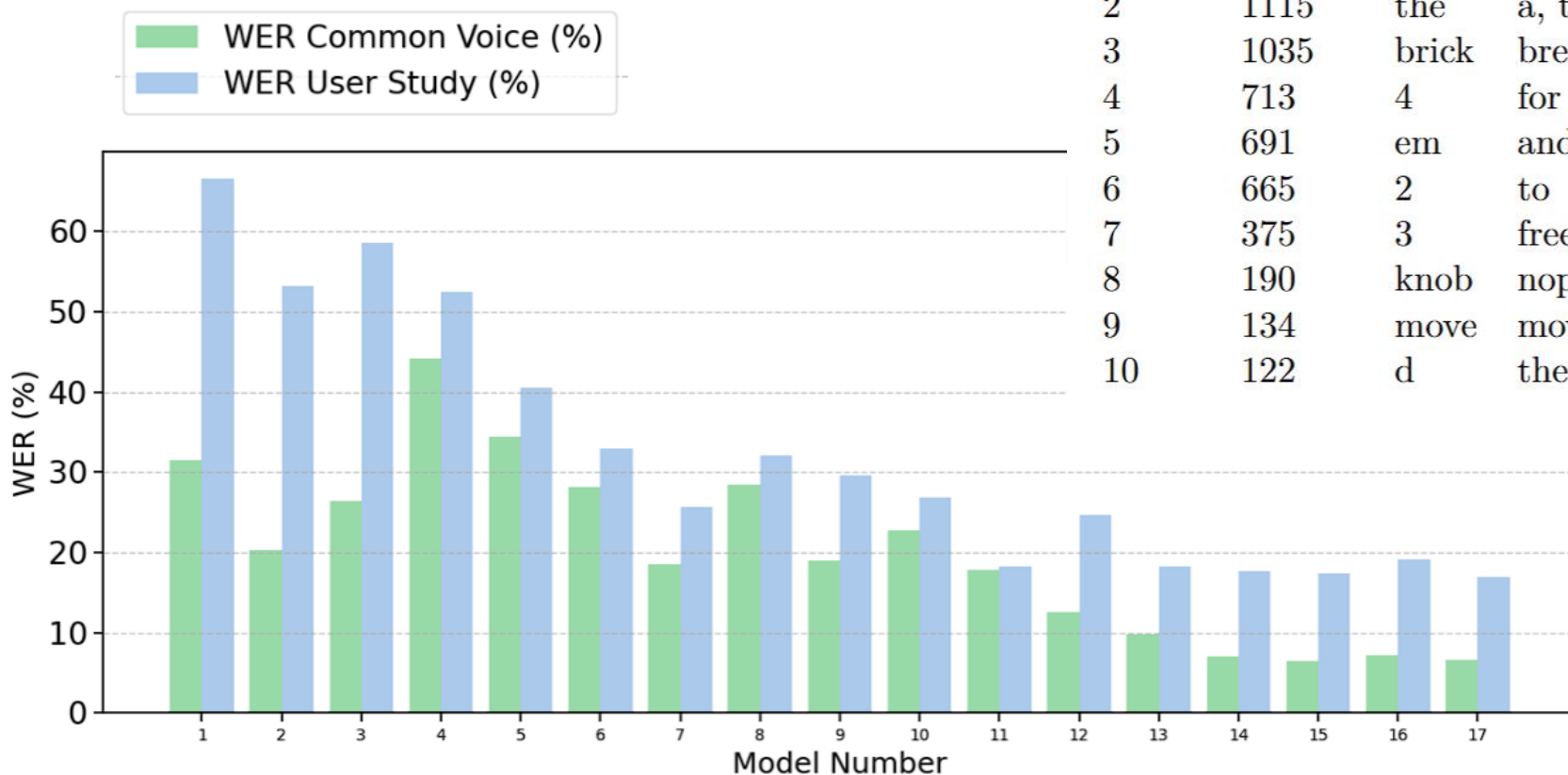
States task instructions or system responses

User Study

- Tasks:
 - Tutorial: Basic commands (place, rotate, color)
 - Task 1: Brick placement with grid annotations
 - Task 2: Stacking bricks
 - Task 3: Chair-building + recoloring



User Study Benchmark



Rank	Count	Ref.	Hyp.
1	2544	by	buy
2	1115	the	a, ther, that, to, de, there, it, te
3	1035	brick	break, brik, bricks, bridge, prick, brig
4	713	4	for
5	691	em	and, m, am
6	665	2	to
7	375	3	free
8	190	knob	nop, knop
9	134	move	moved
10	122	d	the

Statistical Evaluation – Key Findings

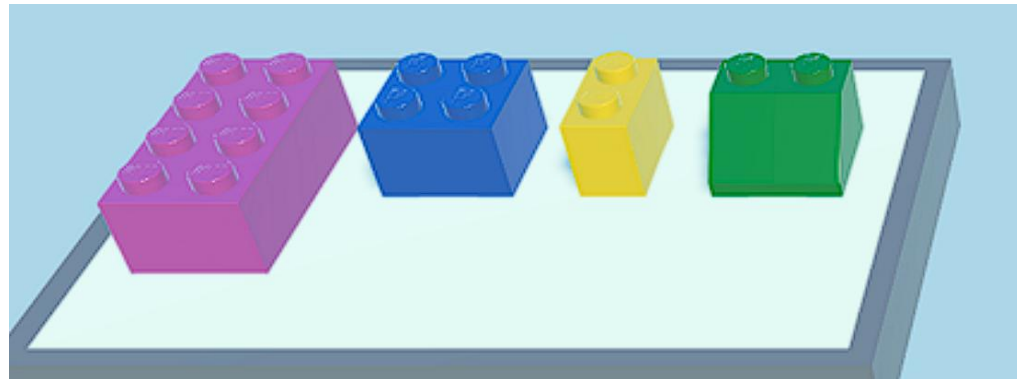
- 33 seconds to position (select, color and place) a brick -> 74% of the time used for thinking, rephrasing and enunciation
- Color selection -> 12% faster task completion

Shape selection {

Take the 2x2 brick

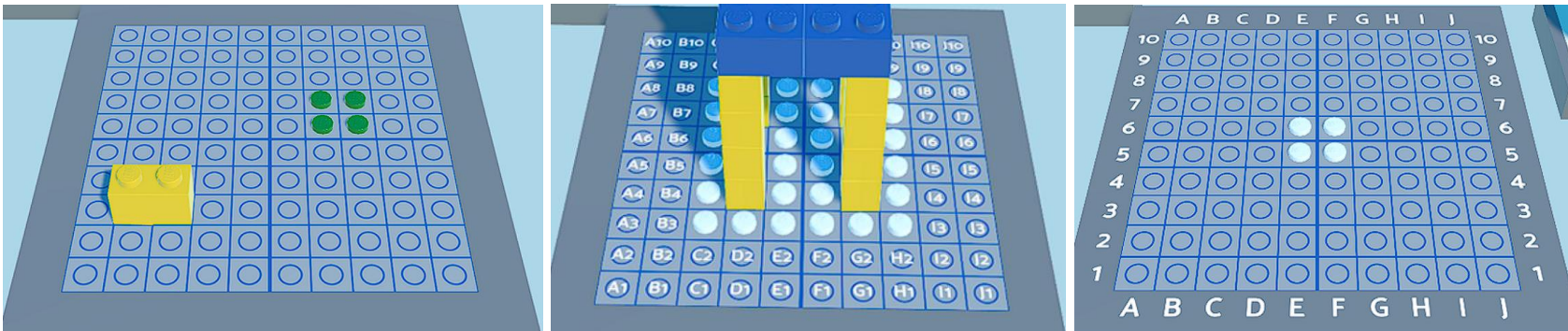
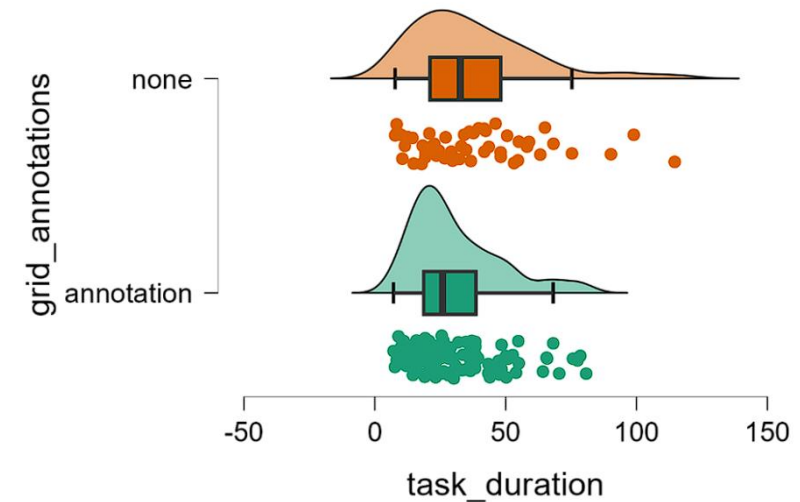
Color selection {

Select the blue brick



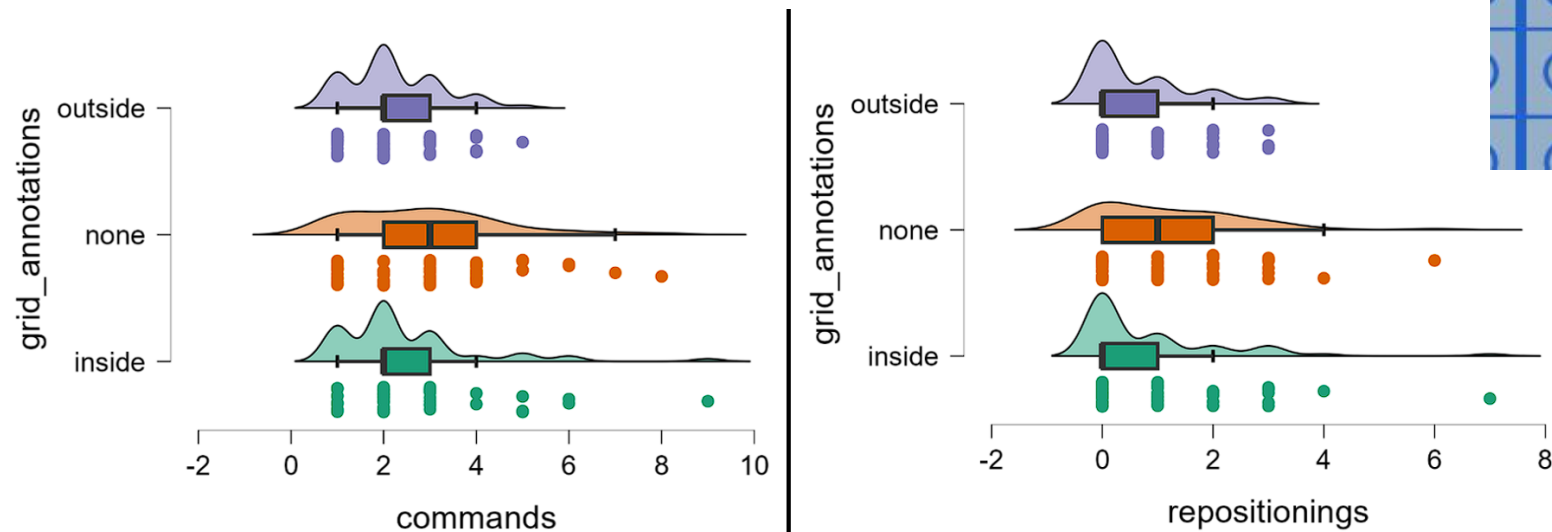
Statistical Evaluation – Key Findings

- 33 seconds to position (select, color and place) a brick -> 74% of the time used for thinking, rephrasing and enunciation
- Color selection -> 12% faster task completion
- Using any grid annotation -> 17% speed increase in task completion

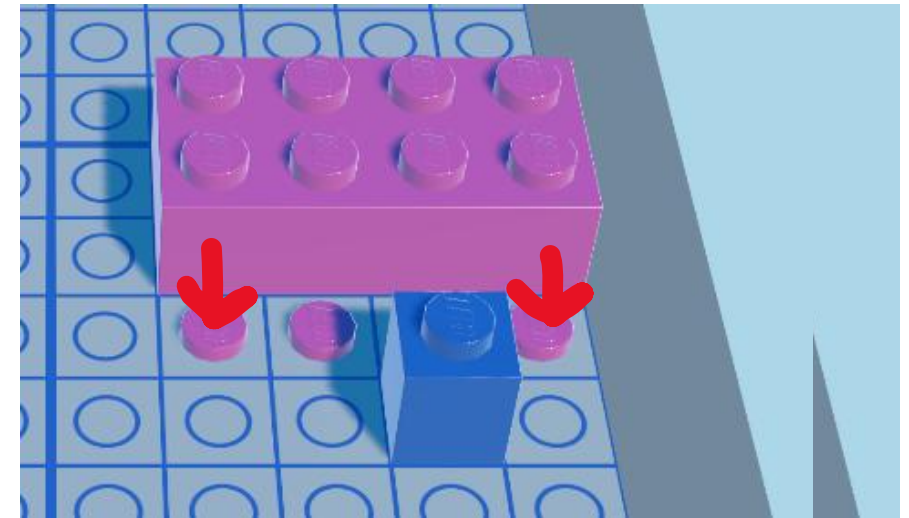


Statistical Evaluation – Key Findings

- Avg. 2.46 commands used to place a brick -> 36% were repositionings
- No repositionings -> 56% speed increase in task completion



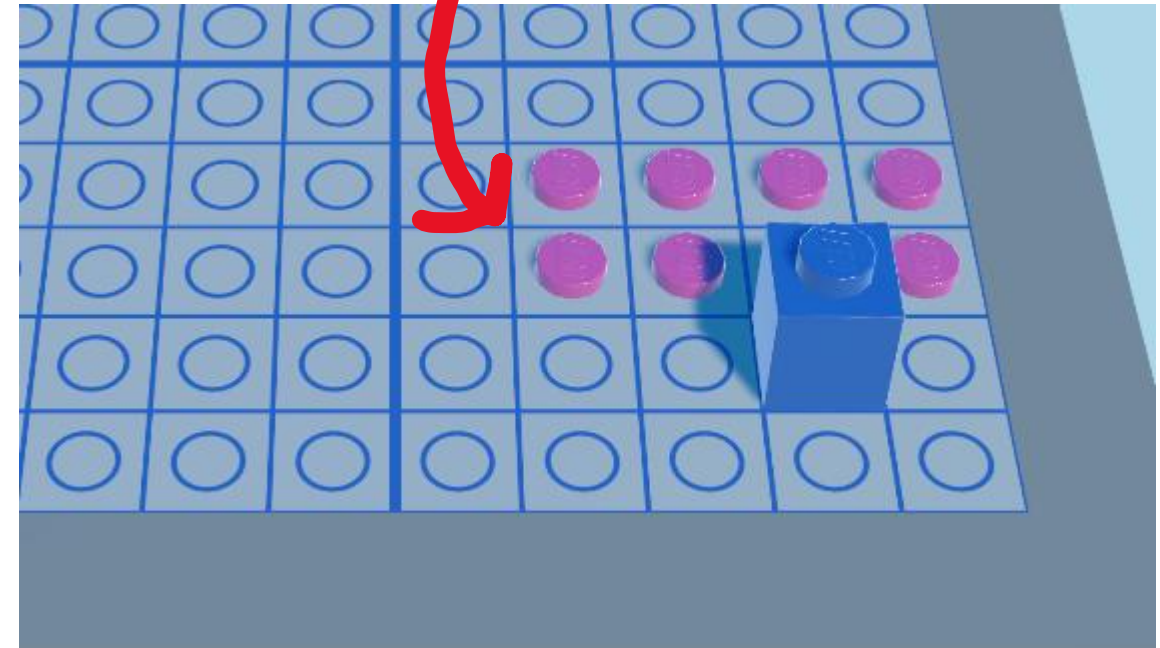
Move it one down!



Statistical Evaluation – Key Findings

- Avg. 2.46 commands used to place a brick -> 36% were repositionings
- No repositionings -> 56% speed increase in task completion
- Using relative positioning -> commands 22% shorter, but 68% more repositionings

Place the pink brick above and next to the blue one



Conclusion & Future Work

- Conclusion:
 - Whisper best for accuracy
 - Wav2Vec for real-time
 - Shape preferred selection metric, color is faster
 - Grid annotations reduce repositionings
- Future Work:
 - Fine-tuning for domain-specific terms (e.g., "brick", "knob").
 - Adaptive interfaces with gaze tracking
 - Streaming for command prediction

Thank you very much to all the participants of the study!
A special thanks to my supervisors, Marcel Ritter and Pascal Knierim!

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User Study Benchmark

- 2.71 hours of audio
- 1311 samples

	Model	Subst. (%)	Del. (%)	Ins. (%)	WER (%)	SER (%)	RTF _x
1	vosk-model-small-en-us-0.15	17.5	45.8	3.2	66.6	93.1	4.01
2	vosk-model-en-us-0.22	16.4	33.3	3.4	53.2	85.7	1.71
3	vosk-model-en-us-0.22-lgraph	21.3	33.2	4.1	58.6	92.1	0.70
4	wav2vec2-base-100h	39.5	6.5	6.5	52.5	98.0	591.22
5	wav2vec2-base-960h	29.0	6.3	5.2	40.5	93.8	636.01
6	wav2vec2-large-960h	22.7	4.6	5.6	32.9	89.9	388.08
7	wav2vec2-large-960h-lv60-self	16.0	5.8	3.9	25.7	79.3	168.54
8	wav2vec2-large-robust-ft-swbd-300h	21.7	5.9	4.4	32.1	89.6	159.27
9	wav2vec2-large-robust-ft-libri-960h	19.8	5.1	4.7	29.6	85.9	171.85
10	whisper-tiny.en	12.1	5.8	8.8	26.8	69.1	81.42
11	whisper-small.en	9.1	6.0	3.1	18.2	61.3	43.03
12	whisper-base.en	10.6	6.0	8.1	24.7	65.5	60.34
13	whisper-medium.en	9.0	6.2	3.0	18.2	61.0	22.00
14	whisper-large	8.8	5.9	3.0	17.7	59.6	10.68
15	whisper-large-v2	8.8	5.7	2.8	17.3	60.3	10.81
16	whisper-large-v3	9.1	5.8	4.1	19.1	60.5	10.60
17	whisper-large-v3-turbo	8.4	5.6	2.9	16.9	59.6	10.52
	Global Best	-	-	-	16.9	59.6	636.01

User Study Benchmark

Rank	whisper-large-v3-turbo			vosk-model-en-us-0.22			wav2vec2-large-960h-lv60-self		
	Count	Ref.	Hyp.	Count	Ref.	Hyp.	Count	Ref.	Hyp.
1	21	brick	break	90	2	to	58	em	m
2	18	em	and	73	brick	break	39	the	te
3	18	it	colored	68	the	to	38	4	for
4	14	4	1	55	by	buy	33	em	and
5	10	the	a	39	4	for	32	brick	prick
6	9	just	is	35	d	the	28	2	to
7	8	brick	bricks	34	3	free	28	3	free
8	7	b	p	32	the	that	24	color	colored
9	7	have	which	31	em	and	24	em	am
10	7	the	that	25	brick	bridge	24	move	moved